LAB 7: FUZZING FOR INPUT VALIDATION BUGS

Hypothesis-based testing and defensive input validation

1. Objective

The objective of this lab was to use the Hypothesis property-based testing library to generate unexpected inputs, identify input-validation bugs in the supplied Python application, and improve the robustness of the application through defensive programming.

2. Tools and Files

Language: Python 3.12.7

Testing tools: pytest and Hypothesis

Application file: processor.py

Test file: test_processor.py

3. Procedure and Results

Step 1 – Install dependencies

Installed the required testing libraries using: python -m pip install hypothesis pytest

Step 2 – Run the original application

The original processor.py was executed with valid sample inputs to confirm its normal behaviour.

```
C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7>python processor.py
==== Running verify_isbn10_format ====
Enter ISBN-10 (e.g. 0-306-40615-2): 0-306-40615-2
Group 306: Starts 0, Checksum 2

==== Running calculate_depreciation_factor ====
Enter age/lifespan in years (e.g. 3/10): 3/10
Depreciation Factor: 0.7

==== Running lookup_condition_multiplier ====
Enter condition (e.g. MINT, GOOD, POOR): GOOD
Multiplier: 0.6

C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7>
```

Step 3 – Write Hypothesis tests

Completed test_processor.py using the @given decorator and the strategy st.text() | st.none(). Each test calls one application function and reports an assertion failure if the function raises an exception.

Step 4 – Run tests against the original application

The initial Hypothesis run reported failures in all three tests. Generated edge cases exposed the following problems:

Function	Problem discovered	Example generated input
verify_isbn10_format	None input, empty/short strings, and invalid first characters caused exceptions.	None, '', '0', ':'

calculate_depreciation_factor	None input, missing separator, and empty numeric fields caused exceptions.	None, '', '/'
lookup_condition_multiplier	None input and unknown/empty conditions caused exceptions.	None, ''

The original run ended with 3 failed tests. The detailed terminal output was captured to demonstrate that Hypothesis found input-validation bugs.

```
C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7>python -m pytest test_processor.py -v
============================================= test session starts =============================================
platform win32 -- Python 3.12.7, pytest-8.3.4, pluggy-1.5.0 -- C:\Users\ridhi\AppData\Local\Programs\Python\Python312\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7
plugins: anyio-4.11.0, hypothesis-6.170.15
collected 3 items

test_processor.py::test_verify_isbn10_format_no_crash FAILED                                          [ 33%]
test_processor.py::test_calculate_depreciation_factor_safe FAILED                                     [ 66%]
test_processor.py::test_lookup_condition_multiplier_safe FAILED                                       [100%]

================================================== FAILURES ===================================================
_____________________________________ test_verify_isbn10_format_no_crash ______________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\runner.py", line 341, in from_call
  |     result: TResult | None = func()
  |                             ^^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\runner.py", line 242, in <lambda>
  |     lambda: runtest_hook(item=item, **kwds), when=when, reraise=reraise
  |             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 182, in _multicall
  |     return outcome.get_result()
  |            ^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_result.py", line 100, in get_result
  |     raise exc.with_traceback(exc.__traceback__)
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\threadexception.py", line 92, in pytest_runtest_call
  |     yield from thread_exception_runtest_hook()
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\threadexception.py", line 68, in thread_exception_runtest_hook
  |     yield
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\unraisableexception.py", line 95, in pytest_runtest_call
  |     yield from unraisable_exception_runtest_hook()
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\unraisableexception.py", line 70, in unraisable_exception_runtest_hook
  |     yield
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
```

```
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\logging.py", line 846, in pytest_runtest_call
  |     yield from self._runtest_for(item, "call")
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\logging.py", line 829, in _runtest_for
  |     yield
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\capture.py", line 880, in pytest_runtest_call
  |     return (yield)
  |             ^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\skipping.py", line 257, in pytest_runtest_call
  |     return (yield)
  |             ^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
  |     res = hook_impl.function(*args)
  |           ^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\runner.py", line 174, in pytest_runtest_call
  |     item.runtest()
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\python.py", line 1627, in runtest
  |     self.ihook.pytest_pyfunc_call(pyfuncitem=self)
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 139, in _multicall
  |     raise exception.with_traceback(exception.__traceback__)
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
  |     res = hook_impl.function(*args)
  |           ^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\python.py", line 159, in pytest_pyfunc_call
  |     result = testfunction(**testargs)
  |              ^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 11, in test_verify_isbn10_format_no_crash
  |     def test_verify_isbn10_format_no_crash(s):
  |                ^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 4 distinct failures. (4 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 13, in test_verify_isbn10_format_no_crash
    |     verify_isbn10_format(s)
    |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\processor.py", line 3, in verify_isbn10_format
    |     first_digit = int(cleaned[0])
    |                   ^^^^^^^^^^^^^^
```

```
    |     first_digit = int(cleaned[0])
    |                   ^^^^^^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ':'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 15, in test_verify_isbn10_format_no_crash
    |     assert False, f"verify_isbn10_format crashed with input {s!r}: {e}"
    | AssertionError: verify_isbn10_format crashed with input ':': invalid literal for int() with base 10: ':'
    | assert False
    | Failing test case: test_verify_isbn10_format_no_crash(
    |     s=':',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 13, in test_verify_isbn10_format_no_crash
    |     verify_isbn10_format(s)
    |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\processor.py", line 5, in verify_isbn10_format
    |     check_char = cleaned[9].upper()
    |                  ~~~~~~~~^^^
    | IndexError: string index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 15, in test_verify_isbn10_format_no_crash
    |     assert False, f"verify_isbn10_format crashed with input {s!r}: {e}"
    | AssertionError: verify_isbn10_format crashed with input '0': string index out of range
    | assert False
    | Failing test case: test_verify_isbn10_format_no_crash(
    |     s='0',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 13, in test_verify_isbn10_format_no_crash
    |     verify_isbn10_format(s)
    |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\processor.py", line 3, in verify_isbn10_format
    |     first_digit = int(cleaned[0])
    |                   ~~~~~~~~^^^
    | IndexError: string index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 15, in test_verify_isbn10_format_no_crash
    |     assert False, f"verify_isbn10_format crashed with input {s!r}: {e}"
    | AssertionError: verify_isbn10_format crashed with input '': string index out of range
    | assert False
    | Failing test case: test_verify_isbn10_format_no_crash(
```

```
    | assert False
    | Failing test case: test_verify_isbn10_format_no_crash(
    |     s='',
    | )
    +---------------- 4 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 13, in test_verify_isbn10_format_no_crash
    |     verify_isbn10_format(s)
    |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\processor.py", line 2, in verify_isbn10_format
    |     cleaned = isbn_str.replace('-', '').strip()
    |               ^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'replace'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 15, in test_verify_isbn10_format_no_crash
    |     assert False, f"verify_isbn10_format crashed with input {s!r}: {e}"
    | AssertionError: verify_isbn10_format crashed with input None: 'NoneType' object has no attribute 'replace'
    | assert False
    | Failing test case: test_verify_isbn10_format_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____ test_calculate_depreciation_factor_safe _____
  + Exception Group Traceback (most recent call last):
    |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\runner.py", line 341, in from_call
    |     result: TResult | None = func()
    |                              ^^^^^^
    |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\runner.py", line 242, in <lambda>
    |     lambda: runtest_hook(item=item, **kwds), when=when, reraise=reraise
    |             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
    |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
    |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
    |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
    |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 182, in _multicall
    |     return outcome.get_result()
    |            ^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_result.py", line 100, in get_result
    |     raise exc.with_traceback(exc.__traceback__)
    |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
    |     teardown.throw(outcome._exception)
    |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\threadexception.py", line 92, in pytest_runtest_call
    |     yield from thread_exception_runtest_hook()
    |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\threadexception.py", line 68, in thread_exception_runtest_hook
    |     yield
    |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
```

```
    yield
  File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
    teardown.throw(outcome._exception)
  File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\unraisableexception.py", line 95, in pytest_runtest_call
    yield from unraisable_exception_runtest_hook()
  File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\unraisableexception.py", line 70, in unraisable_exception_runtes
    yield
  File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
    teardown.throw(outcome._exception)
  File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\logging.py", line 846, in pytest_runtest_call
    yield from self._runtest_for(item, "call")
  File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\logging.py", line 829, in _runtest_for
    yield
  File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
    teardown.throw(outcome._exception)
  File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\capture.py", line 880, in pytest_runtest_call
    return (yield)
            ^^^^^
  File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
    teardown.throw(outcome._exception)
  File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\skipping.py", line 257, in pytest_runtest_call
    return (yield)
            ^^^^^
  File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
    res = hook_impl.function(*args)
          ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\runner.py", line 174, in pytest_runtest_call
    item.runtest()
  File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\python.py", line 1627, in runtest
    self.ihook.pytest_pyfunc_call(pyfuncitem=self)
  File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
    return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
    return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 139, in _multicall
    raise exception.with_traceback(exception.__traceback__)
  File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
    res = hook_impl.function(*args)
          ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\python.py", line 159, in pytest_pyfunc_call
    result = testfunction(**testargs)
             ^^^^^^^^^^^^^^^^^^^^^^^^
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    result = testfunction(**testargs)
             ^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 19, in test_calculate_depreciation_factor_safe
    def test_calculate_depreciation_factor_safe(s):
                ^^^
  File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\hypothesis\core.py", line 2334, in wrapped_test
    raise the_error_hypothesis_found
ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
+-+---------------- 1 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 21, in test_calculate_depreciation_factor_safe
  |     calculate_depreciation_factor(s)
  |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\processor.py", line 10, in calculate_depreciation_factor
  |     age = float(age_str)
  |           ^^^^^^^^^^^^^
  | ValueError: could not convert string to float: ''
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 23, in test_calculate_depreciation_factor_safe
  |     assert False, f"calculate_depreciation_factor crashed with input {s!r}: {e}"
  | AssertionError: calculate_depreciation_factor crashed with input '/': could not convert string to float: ''
  | assert False
  | Failing test case: test_calculate_depreciation_factor_safe(
  |     s='/',
  | )
  +---------------- 2 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 21, in test_calculate_depreciation_factor_safe
  |     calculate_depreciation_factor(s)
  |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\processor.py", line 9, in calculate_depreciation_factor
  |     age_str, life_str = book_age_str.split('/')
  |     ^^^^^^^^^^^^^^^^^
  | ValueError: not enough values to unpack (expected 2, got 1)
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 23, in test_calculate_depreciation_factor_safe
  |     assert False, f"calculate_depreciation_factor crashed with input {s!r}: {e}"
  | AssertionError: calculate_depreciation_factor crashed with input '': not enough values to unpack (expected 2, got 1)
  | assert False
  | Failing test case: test_calculate_depreciation_factor_safe(
  |     s='',  # or any other generated value
  | )
  +---------------- 3 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 21, in test_calculate_depreciation_factor_safe
  |     calculate_depreciation_factor(s)
  |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\processor.py", line 9, in calculate_depreciation_factor
  |     age_str, life_str = book_age_str.split('/')
  |                         ^^^^^^^^^^^^^^^^^^^^^^^
  | AttributeError: 'NoneType' object has no attribute 'split'
  |
  | During handling of the above exception, another exception occurred:
```

```
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 23, in test_calculate_depreciation_factor_safe
    |     assert False, f"calculate_depreciation_factor crashed with input {s!r}: {e}"
    | AssertionError: calculate_depreciation_factor crashed with input None: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_calculate_depreciation_factor_safe(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________________________ test_lookup_condition_multiplier_safe _____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\runner.py", line 341, in from_call
  |     result: TResult | None = func()
  |                             ^^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\runner.py", line 242, in <lambda>
  |     lambda: runtest_hook(item=item, **kwds), when=when, reraise=reraise
  |             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 182, in _multicall
  |     return outcome.get_result()
  |            ^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_result.py", line 100, in get_result
  |     raise exc.with_traceback(exc.__traceback__)
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\threadexception.py", line 92, in pytest_runtest_call
  |     yield from thread_exception_runtest_hook()
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\threadexception.py", line 68, in thread_exception_runtest_hook
  |     yield
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\unraisableexception.py", line 95, in pytest_runtest_call
  |     yield from unraisable_exception_runtest_hook()
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\unraisableexception.py", line 70, in unraisable_exception_runtest_hook
  |     yield
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\logging.py", line 846, in pytest_runtest_call
  |     yield from self._runtest_for(item, "call")
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\logging.py", line 829, in _runtest_for
  |     yield
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\capture.py", line 880, in pytest_runtest_call
  |     return (yield)
  |             ^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
```

```
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\skipping.py", line 257, in pytest_runtest_call
  |     return (yield)
  |             ^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
  |     res = hook_impl.function(*args)
  |           ^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\runner.py", line 174, in pytest_runtest_call
  |     item.runtest()
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\python.py", line 1627, in runtest
  |     self.ihook.pytest_pyfunc_call(pyfuncitem=self)
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 139, in _multicall
  |     raise exception.with_traceback(exception.__traceback__)
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
  |     res = hook_impl.function(*args)
  |           ^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\_pytest\python.py", line 159, in pytest_pyfunc_call
  |     result = testfunction(**testargs)
  |              ^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 27, in test_lookup_condition_multiplier_safe
  |     def test_lookup_condition_multiplier_safe(s):
  |                    ^^^
  |   File "C:\Users\ridhi\AppData\Local\Programs\Python\Python312\Lib\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 29, in test_lookup_condition_multiplier_safe
    |     lookup_condition_multiplier(s)
    |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\processor.py", line 24, in lookup_condition_multiplier
    |     return multipliers[grade]
    |            ~~~~~~~~~~~^^^^^^^
    | KeyError: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 31, in test_lookup_condition_multiplier_safe
    |     assert False, f"lookup_condition_multiplier crashed with input {s!r}: {e}"
    | AssertionError: lookup_condition_multiplier crashed with input '': ''
    | assert False
    | Failing test case: test_lookup_condition_multiplier_safe(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 29, in test_lookup_condition_multiplier_safe
    |     lookup_condition_multiplier(s)
    |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\processor.py", line 16, in lookup_condition_multiplier
    |     grade = condition_code.strip().upper()
    |             ^^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 31, in test_lookup_condition_multiplier_safe
```

```
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7\test_processor.py", line 31, in test_lookup_condition_multiplier_safe
    |     assert False, f"lookup_condition_multiplier crashed with input {s!r}: {e}"
    | AssertionError: lookup_condition_multiplier crashed with input None: 'NoneType' object has no attribute 'strip'
    | assert False
    | Failing test case: test_lookup_condition_multiplier_safe(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
=========================================== short test summary info ============================================
FAILED test_processor.py::test_verify_isbn10_format_no_crash - ExceptionGroup: Hypothesis found 4 distinct failures. (4 sub-exceptions)
FAILED test_processor.py::test_calculate_depreciation_factor_safe - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_processor.py::test_lookup_condition_multiplier_safe - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
============================================ 3 failed in 8.10s =============================================

C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7>
```

Step 5 – Fix the application

Updated processor.py with defensive input checks. The ISBN function checks the input type, cleaned length, and permitted characters before indexing. The depreciation function checks the input type, separator count, numeric conversion, finite values, and valid age/lifespan ranges. The condition lookup checks the input type and uses dict.get() so an unknown condition returns None instead of raising KeyError.

Invalid inputs return None. Valid inputs retain the expected output format and calculation.

Step 6 – Run the fixed application

Executed the updated processor.py with the same valid sample inputs to verify that the application still works for normal use.

The fixed application was run with the same valid sample inputs. The terminal output was:

```
C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7>python processor.py
==== Running verify_isbn10_format ====
Enter ISBN-10 (e.g. 0-306-40615-2): 0-306-40651-2
Group 306: Starts 0, Checksum 2

==== Running calculate_depreciation_factor ====
Enter age/lifespan in years (e.g. 3/10): 3/10
Depreciation Factor: 0.7

==== Running lookup_condition_multiplier ====
Enter condition (e.g. MINT, GOOD, POOR): GOOD
Multiplier: 0.6

C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7>
```

Step 7 – Re-run Hypothesis tests

Re-ran the test suite against the fixed application using python -m pytest test_processor.py -q. All three tests passed.

```
C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7>python -m pytest test_processor.py -q
...
3 passed in 1.14s

C:\Users\ridhi\L6_L7_09_bookstore_trade_in\Lab 7>
```

Screenshot 4: All Hypothesis tests passing (3 passed in 1.14s).

4. Conclusion

This lab demonstrated how property-based fuzz testing can reveal unexpected input validation failures that may not be covered by ordinary example-based tests. Hypothesis generated edge cases for all three functions, and the discovered issues were addressed with defensive validation. The final test run completed successfully with 3 passed tests.

FIXED CODES

processor.py

```
import math


def verify_isbn10_format(isbn_str):
    if not isinstance(isbn_str, str):
        return None

    cleaned = isbn_str.replace('-', '').strip().upper()

    if len(cleaned) != 10:
        return None

    if not cleaned[:9].isdigit():
        return None

    if not (cleaned[9].isdigit() or cleaned[9] == 'X'):
        return None

    first_digit = int(cleaned[0])
    group_code = cleaned[1:4]
    check_char = cleaned[9]

    return f"Group {group_code}: Starts {first_digit}, Checksum {check_char}"



def calculate_depreciation_factor(book_age_str):
    if not isinstance(book_age_str, str):
        return None

    parts = book_age_str.split('/')

    if len(parts) != 2:
        return None
```

```
    try:
        age = float(parts[0])
        life = float(parts[1])
    except ValueError:
        return None

    if not math.isfinite(age) or not math.isfinite(life):
        return None

    if age < 0 or life <= 0:
        return None

    depreciation = 1.0 - (age / life)

    return round(max(0.1, depreciation), 2)



def lookup_condition_multiplier(condition_code):
    if not isinstance(condition_code, str):
        return None

    grade = condition_code.strip().upper()

    multipliers = {
        "MINT": 1.25,
        "LIKE_NEW": 1.00,
        "VERY_GOOD": 0.80,
        "GOOD": 0.60,
        "POOR": 0.30
    }

    return multipliers.get(grade)



def main():
    print("==== Running verify_isbn10_format ====")
```

```
    isbn_in = input("Enter ISBN-10 (e.g. 0-306-40615-2): ")
    print(verify_isbn10_format(isbn_in))

    print("\n==== Running calculate_depreciation_factor ====")
    age_in = input("Enter age/lifespan in years (e.g. 3/10): ")
    print("Depreciation Factor:", calculate_depreciation_factor(age_in))

    print("\n==== Running lookup_condition_multiplier ====")
    cond_in = input("Enter condition (e.g. MINT, GOOD, POOR): ")
    print("Multiplier:", lookup_condition_multiplier(cond_in))



if __name__ == "__main__":
    main()
```

test_processor.py

```
from hypothesis import given, strategies as st
from processor import (
    verify_isbn10_format,
    calculate_depreciation_factor,
    lookup_condition_multiplier
)



@given(st.text() | st.none())
def test_verify_isbn10_format_no_crash(s):
    try:
        verify_isbn10_format(s)
    except Exception as e:
        assert False, f"verify_isbn10_format crashed with input {s!r}: {e}"
```

```
@given(st.text() | st.none())
def test_calculate_depreciation_factor_safe(s):
    try:
        calculate_depreciation_factor(s)
    except Exception as e:
        assert False, f"calculate_depreciation_factor crashed with input {s!r}: {e}"



@given(st.text() | st.none())
def test_lookup_condition_multiplier_safe(s):
    try:
        lookup_condition_multiplier(s)
    except Exception as e:
        assert False, f"lookup_condition_multiplier crashed with input {s!r}: {e}"
```